

MINERAL REGION SEGMENTATION USING MULTISPECTRAL REMOTE SENSING IMAGES BASED ON CNN

By

ANIT KUMAR MAITY

(Registration Number: 24MCA00PY0055)

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SCHOOL OF ENGINEERING & TECHNOLOGY
PONDICHERY UNIVERSITY
PUDUCHERRY -605014**

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BONAFIDE CERTIFICATE

This is to certify that this project work entitled 'Mineral Region Segmentation Using Multispectral Remote Sensing Images Based on CNN' is a Bonafide record of work done by **Mr. Anit Kumar Maity** (Reg. Number **24MCA00PY0055**) in the partial fulfilment for the Degree of **Master of Computer Applications** in the **Department of Computer Science, School of Engineering and Technology** of **Pondicherry University**.

This work has not been submitted elsewhere for the award of any other degree to the best of our knowledge.


INTERNAL GUIDE

Dr. V. UMA

Associate Professor

Department of Computer Science

School of Engineering & Technology

Pondicherry University

Pondicherry – 605 014

HEAD OF THE DEPARTMENT

Dr. S. K. V. JAYAKUMAR

Professor & HOD

Department of Computer Science

School of Engineering & Technology

Pondicherry University

Pondicherry – 605 014

Submitted for the Viva-Voce Examination held on:


INTERNAL EXAMINER


EXTERNAL EXAMINER

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TABLE OF CONTENTS

TITLE	PAGE NO
ACKNOWLEDGEMENT	
1. ABSTRACT	1
2. INTRODUCTION	2
3. LITERATURE REVIEW	3
4. MOTIVATION & PROBLEM STATEMENT	4
4.1 MOTIVATION	
4.2 PROBLEM STATEMENT	
4.3 OBJECTIVES	
4.3.1 DATA ACQUISITION AND PREPROCESSING	
4.3.2 MODEL DEVELOPMENT AND TRAINING STRATEGY	
4.3.3 PERFORMANCE EVALUATION AND COMPARATIVE ANALYSIS	
5. METHODOLOGY	6
5.1 OVERVIEW OF THE PROPOSED FRAMEWORKS	
5.2 MODEL ARCHITECTURES	
5.2.1 PROPOSED UNET ARCHITECTURE	
5.2.2 PROPOSED ATTENTION UNET ARCHITECTURE	
5.2.3 PROPOSED RESUNET ARCHITECTURE	
5.3 MODEL TRAINING	
5.4 SYSTEM DETAILS	
6. RESULTS & DISCUSSION	20
6.1 ABLATION STUDY	
6.2 MODEL COMPARISON	
6.3 DISCUSSION	
6.4 SPATIAL ANALYSIS	
6.5 KEY FINDINGS	
7. CONCLUSION & FUTURE ENHANCEMENT	25
8. REFERENCES	26

LIST OF FIGURES

TITLE	PAGE NO
5.1: WORKFLOW OF THE PROPOSED MINERALISATION SEGMENTATION SYSTEM	6
5.2: STUDY AREA (SALEM REGION)	7
5.3: RAW 7-BAND INPUT (B1 – B7) AND RGB COMPOSITE	8
5.4: PCA-DERIVED 3-COMPONENT INPUT (PC1, PC2, PC3)	9
5.5: LITHOLOGIC, INTRUSIVE, AGE, GROUP_NAME(GEOLOGY) AND DISTANCE-TO-FAULT	10
5.6: GROUND TRUTH DATA FOR THE SALEM REGION	11
5.7: PATCH SPLIT SUMMARY PRE-AUGMENTATION AND AFTER AUGMENTATION	12
5.8: U-NET ARCHITECTURE DIAGRAM	14
5.9: ATTENTION U-NET ARCHITECTURE DIAGRAM	16
5.10: RESUNET ARCHITECTURE DIAGRAM	18
6.1: U-NET TRAINING AND VALIDATION CURVES AND CONFUSION MATRICES	21
6.2: ATTENTION U-NET TRAINING AND VALIDATION CURVES AND CONFUSION MATRICES	22
6.3: RESUNET TRAINING AND VALIDATION CURVES AND CONFUSION MATRICES	23
6.4: MINERAL PROSPECTIVITY MAP	24

LIST OF TABLES

TITLE	PAGE NO
5.1: PCA VARIANCE TABLE	9
6.1: PERFORMANCE COMPARISON TABLE	20

CHAPTER 01

ABSTRACT

Mineral prospectivity mapping (MPM) always plays a vital role in recognizing prospective mineral region for sustainable resource exploration. Traditional approaches, although consistent, are frequently laborious, expensive, and restricted in spatial coverage. To address these challenges, this study offers a deep learning-based framework for mineral prospectivity mapping in the Salem region, Tamil Nadu, which is known for its complex geological settings and minerals ore potential. In this research, Multispectral data from Landsat_8 along with geological and structural data are used to improve prediction accuracy. Additionally, Principal Component Analysis (PCA) is applied to decrease dimensionality and capture the main spectral features. A patch-based dataset is formed using a sliding window method, followed by data preprocessing, class balancing, and data augmentation to address insufficiency and class unequally. Three model variants are trained using Weighted BCE + Dice loss (pos_weight = 45) and the AdamW optimiser with early stopping and validation-monitored checkpointing, all evaluated at a fixed decision threshold of 0.30: UNet-PCA3 (3-channel PCA input, deep supervision, 92 epochs) achieved a test Mineral IoU of 90.67% and F1-Score of 95.11%; AttentionUNet-8ch (3 PCA + 5 geological feature channels, attention gates, 120 epochs) achieved a test Mineral IoU of 88.84% and F1-Score of 94.09%; and ResUNet-15ch (7 raw bands + 3 PCA + 5 geological feature channels, residual blocks, 120 epochs) achieved a test Mineral IoU of 88.51% and F1-Score of 93.90%, representing its efficiency in capturing complex spatial and spectral patterns for creating mineral mapping. The proposed method highlights the importance of multi-source data amalgamation and advanced deep learning architectures for improved mineral prediction accuracy. This study is extremely useful for geologists, as it offers reliable mineral potential maps that can guide field surveys, decrease time and cost, and support focus on high-probability zones. This will also contribute to the development of effective and reliable mineral exploration systems to support sustainable resource development by aiding better identification of potential mineral zones.

CHAPTER 02

INTRODUCTION

Scene understanding from satellite remote sensing imagery is essential for geological mapping, mineral exploration, and land cover analysis. Accurately identifying mineralised zones from large-scale satellite data supports efficient field investigations, reduces exploration costs, and enables informed decision-making in regions that are difficult to access. However, traditional techniques such as spectral indices, band combinations, and Principal Component Analysis (PCA) often struggle when mineral signatures overlap with background classes. These methods also fail to capture spatial relationships between neighbouring pixels and are sensitive to noise, seasonal variation, and spectral mixing, leading to inconsistent results.

Deep learning approaches, particularly Convolutional Neural Networks (CNNs), have significantly improved segmentation performance. Architectures like U-Net use an encoder–decoder framework with skip connections to preserve spatial details, making them suitable for pixel-level classification. Attention U-Net enhances this by focusing on relevant regions, improving detection of sparse and irregular mineral targets. ResUNet further addresses training challenges in deeper networks through residual connections, enabling stable learning.

In this project, all three architectures — U-Net, Attention U-Net, and ResUNet — are adopted and comparatively evaluated for binary mineralisation mapping using Landsat 8 Collection 2 Level-2 multispectral imagery (7 bands, 30 m resolution) over Salem District, Tamil Nadu, India, a geologically active region with known occurrences of magnetite, hematite, and associated alteration minerals. Three model variants are evaluated across progressively richer feature representations: UNet-PCA3, using the top three PCA-derived components as input; AttentionUNet-8ch, augmenting PCA features with five co-registered geological raster layers encoding lithological unit, intrusive contact, geological age, stratigraphic group, and distance to fault; and ResUNet-15ch, combining all seven raw surface reflectance bands with both PCA components and the same five geological rasters as a unified 15-channel input. The preprocessing module covers the complete pipeline from data acquisition, cloud masking, spectral index and PCA derivation, geological feature integration, and ground truth generation through to model training, forming the foundation for comparative architectural evaluation across the three segmentation frameworks.

CHAPTER 03

LITERATURE REVIEW

Mineral prospectivity mapping (MPM) has evolved significantly from traditional field-based methods to advanced data-driven approaches. Early exploration relied on geological mapping, geochemical analysis (Hawkes and Webb, 1962), and geophysical surveys (Telford et al., 1990) to identify mineralised zones. While valuable, these methods were time-consuming, costly, and spatially limited.

The advent of satellite remote sensing marked a turning point. Crosta and Moore (1989) demonstrated PCA for hydrothermal alteration mapping, while Loughlin (1991) applied band ratio methods for iron oxide and clay mineral identification. Landsat and ASTER systems enabled regional-scale lithological mapping, reducing dependence on fieldwork. Integration with GIS further strengthened MPM; Bonham-Carter (1994) introduced the weights-of-evidence method, and Carranza (2008) confirmed GIS-based spatial modelling for mineral potential detection.

Machine learning (ML) subsequently addressed limitations of GIS-based methods. Random Forest, SVM, and Artificial Neural Networks demonstrated improved prediction accuracy over traditional statistical approaches (Rodríguez-Galiano et al., 2015; Cracknell and Reading, 2014). However, challenges including class imbalance, overfitting, and parameter tuning remained critical concerns (Shirmard et al.; Senanayake et al., 2023).

Deep learning (DL) has recently advanced MPM further, with CNNs automatically extracting complex spatial features from high-dimensional datasets (Kang Sun et al.). Despite superior performance, DL methods face challenges including limited labelled data, high computational costs, and poor model interpretability. Region-specific applications, such as Salem, Tamil Nadu, remain underexplored.

Overall, the evolution of mineral prospectivity mapping demonstrates a clear transition from traditional field-based methods to advanced machine learning and deep learning approaches. While traditional methods provide fundamental geological understanding, modern techniques integrate remote sensing, GIS, and computational models to enable efficient and large-scale mineral exploration. However, there is still a need for integrated frameworks that combine multi-source geospatial data with optimized machine learning models to improve prediction accuracy and reduce uncertainty in mineral prospectivity mapping.

CHAPTER 04

MOTIVATION AND PROBLEM STATEMENT

4.1 MOTIVATION

- Salem District, Tamil Nadu, hosts significant occurrences of magnetite and hematite, yet large-scale automated mineralisation mapping remains limited due to conventional methods being time-consuming, costly, and spatially restricted.
- Traditional remote sensing techniques (spectral indices, band ratios, PCA) fail to capture complex spatial relationships and produce inconsistent results under spectral overlap and extreme class imbalance (2,509:1 background-to-mineral ratio).
- Free, high-resolution Landsat-8 multispectral imagery (7 bands, 30 m) combined with auxiliary geological datasets provides a strong foundation for a cost-effective, data-driven exploration framework.
- Despite proven success in pixel-level segmentation, deep learning encoder-decoder architectures (UNet, Attention UNet, ResUNet) remain largely unapplied to region-specific mineral prospectivity mapping under severely data-scarce and imbalanced conditions, motivating this comparative study.

4.2 PROBLEM STATEMENT

This project addresses the automatic segmentation of mineralised zones from Landsat 8 multispectral imagery of Salem District, Tamil Nadu, through a comparative evaluation of three deep learning encoder-decoder frameworks: a standard U-Net, an Attention U-Net, and a Residual U-Net (ResUNet).

Input Data:

- Landsat 8 Collection 2 Level-2 surface reflectance imagery (7 bands: B1–B7, 30 m resolution), scene LC08_L2SP_143052_20230213_20230218_02_T1
- All seven raw spectral bands as full input (RawBands7ch)
- Top three PCA-derived components as a compact 3-channel spectral representation
- Five co-registered geological raster layers: lithological unit, intrusive contact flag, geological age, stratigraphic group, and distance to nearest fault
- Binary ground truth mask generated from a mineralisation shapefile using a 90 m buffer and rasterisation aligned to the Landsat grid

Output Data:

- Binary segmentation map (mineralised vs. background) at 30 m resolution

- Evaluation measures: Mineral IoU, Mean IoU, F1-Score, Precision, and Recall on held-out test patches

Challenges:

- Extreme class imbalance ($\approx 0.04\%$ mineral pixels; $\sim 2509:1$ background-to-mineral ratio).
- Limited labelled data requiring buffer-based ground truth and aggressive augmentation ($10\times$ mineral, $3\times$ background).
- Comparative evaluation across three architectures and three feature channel configurations (7-channel Raw Bands, 3-channel PCA, 12-channel RAW+Geo, 8-channel PCA+Geo, 15-channel Raw+PCA+Geo).
- Strict spatial patch splitting with 50% overlap to prevent data leakage across train, validation, and test sets.

4.3 OBJECTIVES

4.3.1 Data Acquisition and Preprocessing:

- Acquire and preprocess Landsat 8 Level-2 imagery, including radiometric scaling, QA_PIXEL cloud masking, and spatial sub-setting to the Salem study region.
- Perform PCA on the seven-band stack and extract the top three components as a dimensionality-reduced spectral input.
- Integrate five auxiliary geological raster features — lithology, intrusive contacts, geological age, stratigraphic group, and distance to fault — as supplementary input channels.
- Generate a buffered (90 m) and rasterised binary mineral ground truth mask from the geological mineralisation shapefile.

4.3.2 Model Development and Training Strategy:

- Implement and train three segmentation architectures — UNet-PCA3, AttentionUNet-8ch and ResUNet-15ch — using all possible channel combinations, with a Weighted Binary Cross-Entropy (BCE) + Dice loss to handle class imbalance.

4.3.3 Performance Evaluation and Comparative Analysis:

- Evaluate and compare all three models using pixel-wise Mineral IoU, F1-Score, Precision, and Recall on the held-out test split to identify the optimal architecture and feature representation for mineralisation mapping.

CHAPTER 05

METHODOLOGY

5.1 OVERVIEW OF THE PROPOSED FRAMEWORK

An end-to-end mineralisation segmentation system is proposed that processes Landsat 8 multispectral imagery through a structured pipeline of five stages: satellite imagery acquisition, data preprocessing, feature engineering across three representations, parallel deep learning model training under three architectures, and binary map output with evaluation. Figure 5.1 illustrates the complete workflow.

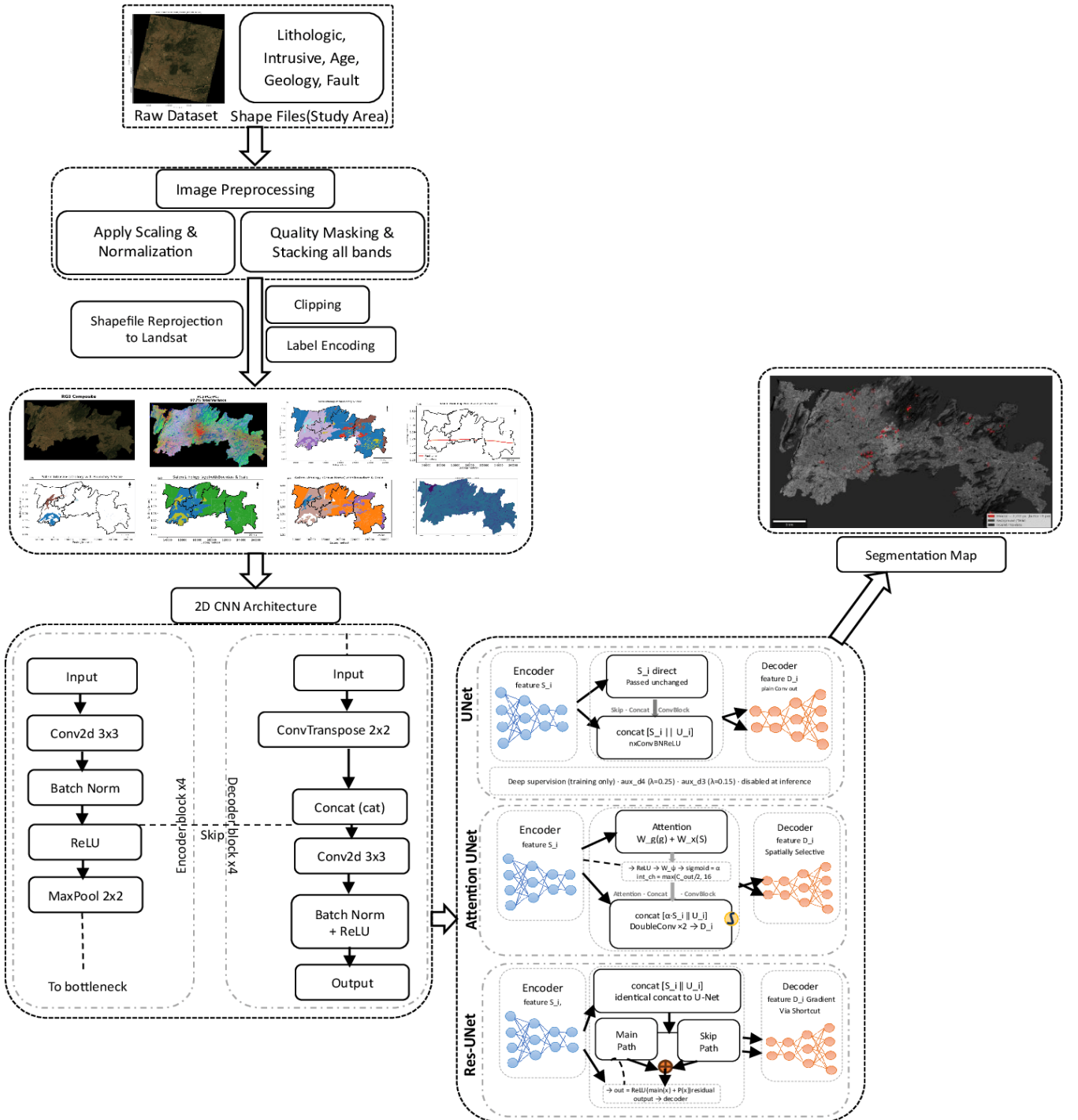


Figure 5.1 Workflow of the proposed mineralisation segmentation system

5.1.1 Study Area and Data Acquisition:

Landsat 8 Collection 2 Level-2 surface reflectance imagery is acquired from the USGS EarthExplorer portal for scene path/row 143/052 (Scene ID: LC08_L2SP_143052_20230213_20230218_02_T1, dated 13 February 2023). The study area covers Salem District, Tamil Nadu, India — a geologically significant region with documented occurrences of magnetite, hematite, and associated alteration minerals. Five sub districts are targeted: Attur, Mettur, Omalur, Salem, and Sankari, covering approximately $120 \text{ km} \times 100 \text{ km}$ at 30 m pixel resolution. The geographical location and spatial coverage of the study area are presented in Figure 5.2.

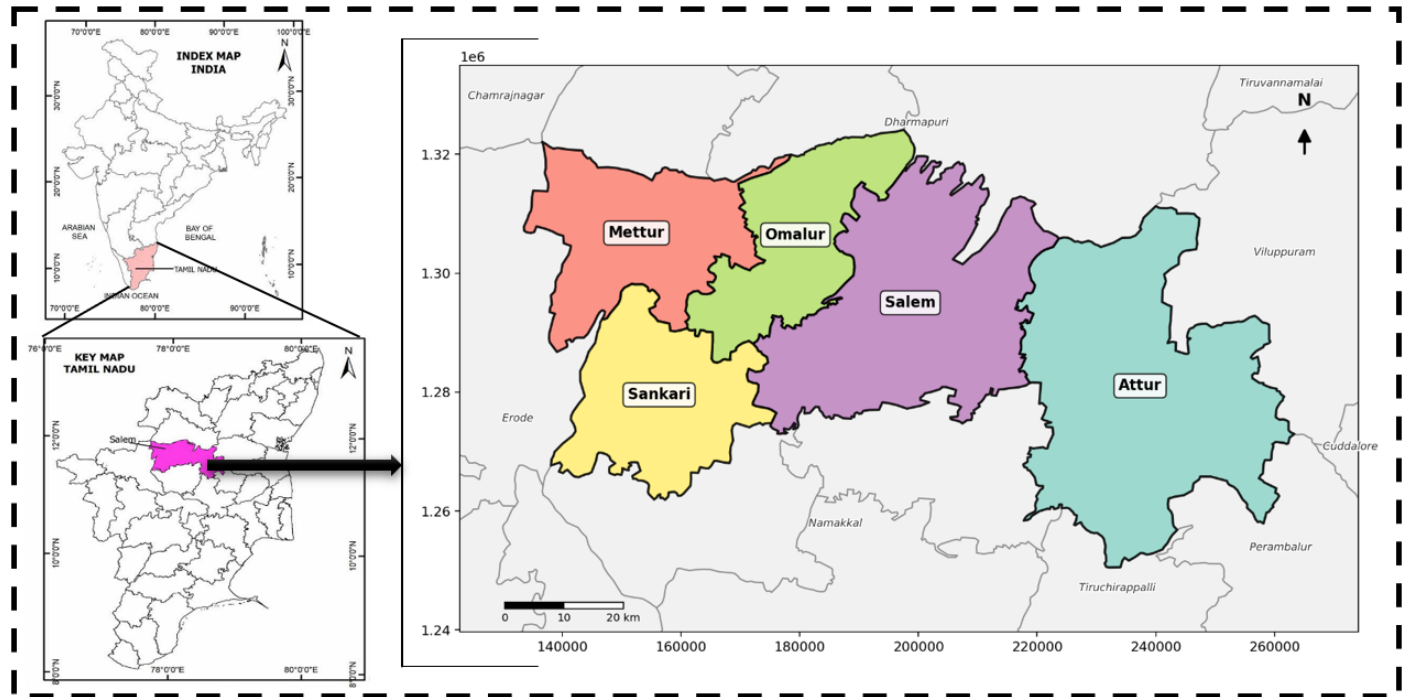


Figure 5.2 Study area (Salem region)

5.1.2 Data Preprocessing:

The preprocessing pipeline involves four sequential steps applied to the Landsat 8 Level-2 surface reflectance data corresponding to the study region.

- **Band loading:** All seven surface reflectance bands (B1–B7) are loaded individually from their respective GeoTIFF files. The full Landsat scene dimensions are $7,791 \times 7,641$ pixels at 30 m resolution, in CRS EPSG:32644 (WGS 84 / UTM Zone 44N), spanning Easting 109,785–339,015m and Northing 1,162,785–1,396,515 m.
- **Radiometric scaling:** Raw DN values are converted to surface reflectance using the standard Landsat Collection 2 formula: $\text{Reflectance} = \text{DN} \times 0.0000275 - 0.2$, with the result clipped to $[0, 1]$. This is applied on load for every band before any further processing.
- **Cloud and shadow masking:** The QA_PIXEL band is decoded using bitwise operations to extract cloud (bit 3), cloud shadow (bit 4), and dilated cloud (bit 1) flags. Pixels flagged on any of these

bits are set to NaN across all seven bands. Applied over the full scene, 59,286,750 pixels (99.6%) are retained as valid, confirming the 13 February 2023 acquisition is effectively cloud-free.

- **Spatial subsetting:** The GADM Level-3 administrative shapefile (gadm41_IND_3.shp, 2,347 features, EPSG:4326) is filtered to the five target sub-districts — Attur, Mettur, Omalur, Salem, and Sankari — then reprojected to EPSG:32644 to match the Landsat CRS. The merged Salem union geometry is used to crop all seven bands via rasterio.mask (crop=True, filled=False), producing a cropped array of shape (7, 2455, 4327), corresponding to a coverage of 73.7 km × 129.8 km ($\approx 9,561$ km²). The QA-derived mask is re-applied after cropping, yielding 5,794,980 valid pixels (54.6% of the cropped extent). The processed array is saved as salem_multispectral_cropped_8.npy along with the boolean valid mask (salem_valid_mask_8.npy), a GeoTIFF (salem_multispectral_cropped_8.tif), and spatial metadata in salem_metadata_8.json.

5.1.3 Feature Engineering:

Four independent feature representations are prepared, each used separately to train all three distinct models:

- **Raw 7-Band Input:** All seven surface reflectance bands (B1–B7) are used directly as the input feature stack (7-channel, 128×128 per patch). This representation gives the model complete access to the original spectral information without any transformation, allowing it to learn band-specific mineral signatures end-to-end. Per-band Z-score normalisation is applied using statistics computed from training patches only. A schematic representation of the raw 7-band input configuration is shown in Figure 5.3.

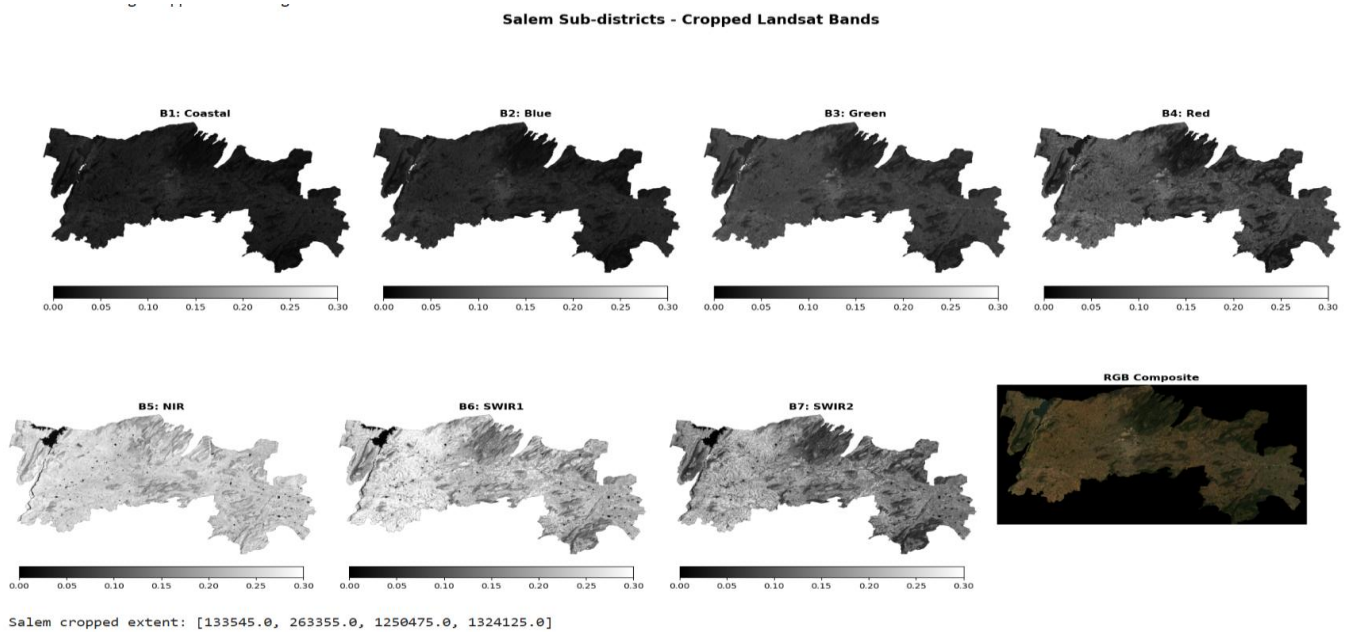


Figure 5.3: Raw 7-Band Input (B1 – B7) and RGB composite

- **PCA-derived 3-component input:** The cropped seven-band array is reshaped from (7, 2455, 4327) into a pixel matrix of shape (10,622,785, 7) and restricted to the 5,794,980 valid (non-NaN) rows.

The valid pixel matrix is standardised using StandardScaler (zero mean, unit variance) and full seven-component PCA is computed. The explained variance distribution is shown in Table 5.1.

Component	Variance	Variance %	Cumulative %
PC1	5.5962	79.95%	79.95%
PC2	0.9008	12.87%	92.82%
PC3	0.3408	4.87%	97.68%
PC4	0.1260	1.80%	99.48%
PC5	0.0152	0.22%	99.70%
PC6	0.0133	0.19%	99.89%
PC7	0.0076	0.11%	100.00%

Table 5.1: PCA Variance Table

PC1 = 79.95%, PC2 = 12.87%, PC3 = 4.87%, PC4 = 1.80%, PC5 = 0.22%, PC6 = 0.19%, with the first three components cumulatively capturing 97.68% of total spectral variance. Accordingly, only the top three components are retained. PC1 encodes overall brightness and soil reflectance, PC2 captures vegetation versus bare-ground contrast, and PC3 encodes subtle mineralogical variation relevant to iron-oxide and clay-bearing zones. The three components are saved both as a compressed archive (salem_pca_top3_bands_8.npz, including component vectors, scaler mean and scale) and as individual georeferenced GeoTIFFs — salem_PC1.tif, salem_PC2.tif, salem_PC3.tif — spatially aligned to the cropped Landsat grid at EPSG:32644. The spatial distribution of the top three principal components is illustrated in Figure 5.4.

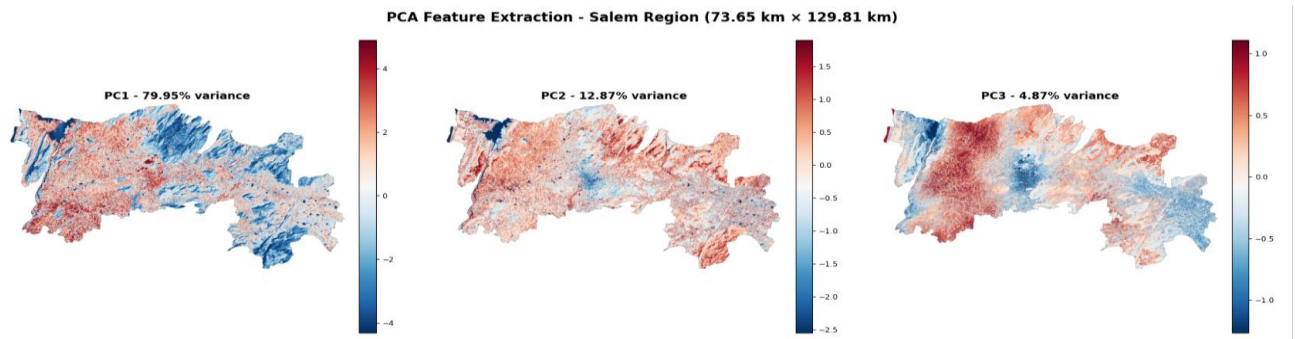


Figure 5.4: PCA-Derived 3-Component Input (PC1, PC2, PC3)

- Geological Feature Preparation (Lithology and Distance-to-Fault Layers):** A regional lithology shapefile (Lithology.shp), originally containing 38,256 polygons in EPSG:4326, is reprojected to EPSG:32644 and clipped to the Salem boundary, reducing it to 1,238 polygons representing 34 lithological units. Four key attributes—LITHOLOGIC, INTRUSIVE, AGE, and GROUP_NAME—are converted into integer codes using a safe encoding method. In this method, missing values (NaN) are assigned a code of 0, while valid categories are assigned positive integers (1, 2, 3, ...). Specifically, LITHOLOGIC has 34 categories (codes 1–34, no missing values), INTRUSIVE has 5 categories (codes 0–5, with some missing values), AGE has 6 categories (codes 1–6), and GROUP_NAME has 8 categories (codes 0–8, including some missing values). Each

encoded layer is converted into raster format using rasterization and saved as separate GeoTIFF files. All rasters are aligned with the Landsat grid (2455×4327) and shared the same coordinate system (EPSG:32644) and spatial properties as other input layers.

In addition, a fault line shapefile (FAULT_2M.shp), originally containing 15 features, is reprojected and clipped to the same study area, resulting in one fault feature. This fault is converted into a binary raster mask, where fault pixels are assigned a value of 1 and all other areas are 0. A distance transform is then applied to calculate the distance of each pixel from the nearest fault. These distances are converted from pixels to metres using the 30 m resolution of the data and then normalised to a range between 0 and 1. The final output, saved as DISTANCE_TO_FAULT.tif, has the same spatial resolution and coordinate system as the other feature layers, ensuring consistency across all inputs used in the model. The spatial distribution of the rasterised geological features and the normalised distance-to-fault map are depicted in Figure 5.5.

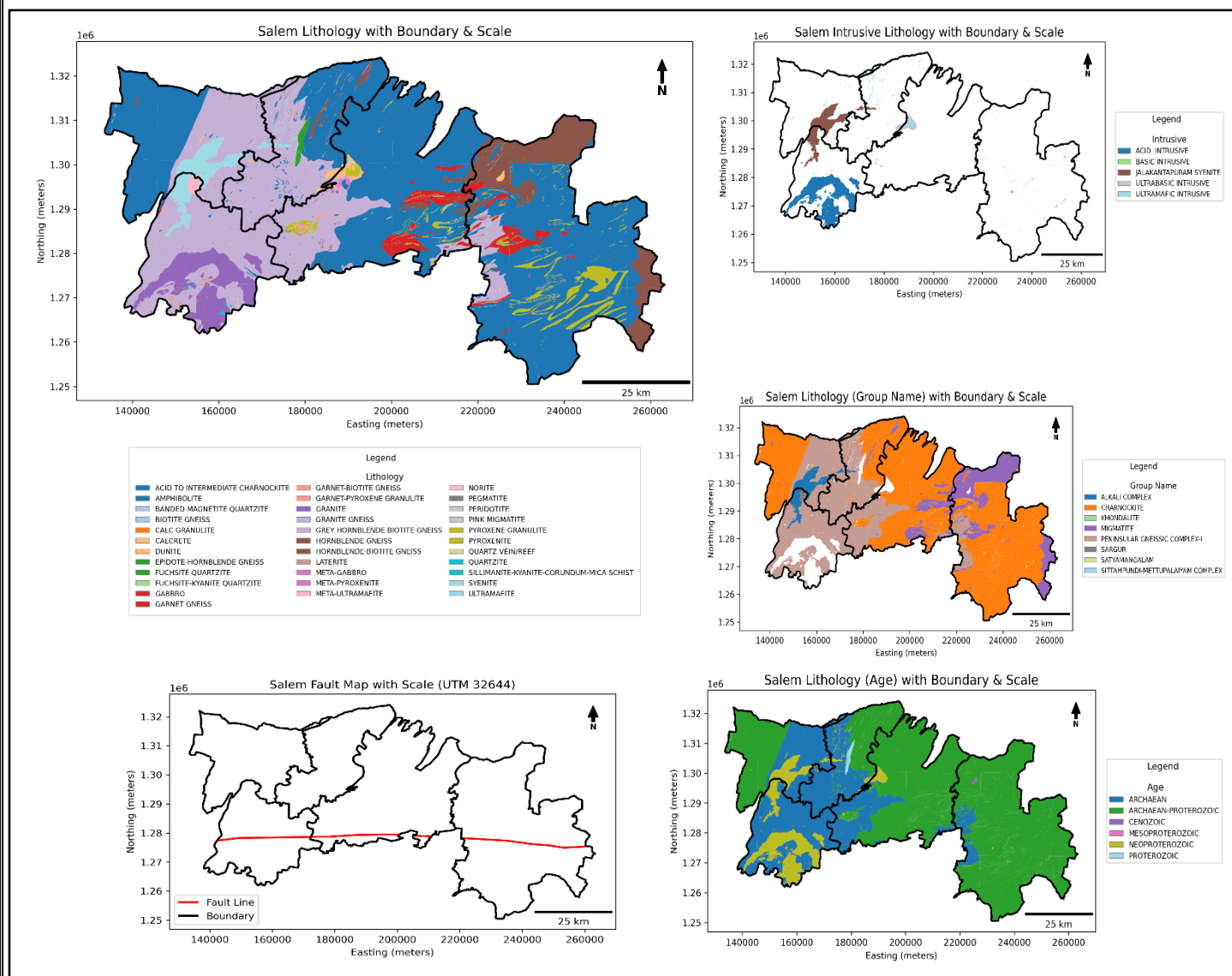


Figure 5.5: Lithologic, Intrusive, Age, Group_Name(Geology) and Distance-To-fault

The final feature inventory comprises 15 channels in total: 7 raw surface reflectance bands, 3 PCA components, and 5 geological rasters (Lithology, Intrusive, Age, Group Name, Distance to Fault). These are combined into five input configurations — 7-channel Raw-bands, 3-channel PCA, 12-channel Raw+Geo, 8-channel PCA+Geo, and 15-channel Raw+PCA+Geo — corresponding to the three model variants.

5.1.4 Ground Truth Generation:

A geological mineralisation point shapefile (Mineralization.shp, EPSG:4326) containing known mineral occurrence locations is used to produce a binary ground truth raster. The procedure is:

1. Shapefile reprojected from EPSG:4326 to the Landsat UTM CRS (EPSG:32644) to ensure metric spatial correspondence.
2. A 90 m spatial buffer ($\text{pixel_size} \times 3 = 30 \times 3$) applied around each mineral occurrence geometry to account for positional uncertainty in the point data and to ensure sufficient pixel labelling density at 30 m resolution.
3. Buffered geometries rasterised onto the full Landsat scene grid ($7,791 \times 7,641$) using `rasterio.features.rasterize()` with `all_touched=True` and `fill=0`, producing a scene-wide binary mask saved as `Minerals_GT_Mask_8.tif`.
4. The scene-wide mask is cropped to the Salem boundary using the same `rasterio.mask` procedure applied during preprocessing, producing `Salem_Minerals_GT_Mask_8.tif` of shape (2455, 4327), with spatial alignment to the Landsat bands verified by confirming identical shape, affine transform, and CRS.
5. The cropped mask was thresholded (`gt_mask > 0`) to produce a uint8 binary label array, which is saved as both `salem_Minerals_binary_labels_8.npy` and `salem_Minerals_binary_labels_8.tif`. The generated binary ground truth mask for mineralisation is illustrated in Figure 5.6.



Figure 5.6: Ground truth data for the Salem region

Of the 5,794,980 valid pixels within the Salem extent, only 2,309 carry a positive mineral label (0.0398%), with the remaining 5,792,671 classed as background — corresponding to a background-to-mineral ratio of approximately 2,509:1, confirming the extreme class imbalance that drives all subsequent loss weighting and augmentation decisions.

5.1.5 Patch Extraction and Augmentation:

Training data is prepared by extracting 128×128-pixel patches using a sliding window with stride 64 and a maximum NaN threshold of 30%. A patch is classified as mineral if at least 0.1% of its pixels are labelled mineral, otherwise as background. From 2,442 total candidate patches, 1,126 are discarded due to excessive NaN coverage, yielding 127 mineral patches and 1,189 background patches. To address class imbalance, mineral patches are augmented 10 times and background patches 3 times during training only (using Albumentations library performed image augmentation: horizontal and vertical flips, random rotation, and brightness/contrast jitter, all at p=0.5), producing final DataLoader splits of 1,672 training, 57 validation, and 60 test patches. A zero-overlap spatial block split is strictly enforced — no patch origin is shared between any two splits — to prevent data leakage across all three models and feature configurations. A visual representation of the patch augmentation process and its resulting outputs is shown in Figure 5.7.

PATCH EXTRACTION – Non-overlapping stride (zero leakage)		Computing per-channel stats from 264 ORIGINAL train patches...	
Patch size	: 128×128	PC1	mean = 0.2430 std = 2.3344
Stride	: 64	PC2	mean = 0.0202 std = 0.8500
Min mineral frac	: 0.001	PC3	mean = 0.0276 std = 0.5917
Mineral aug	: 10× per patch (train only)	Lithology	mean = 10.0779 std = 9.1738
Background aug	: 3× per patch (train only)	Intrusive	mean = 0.2108 std = 0.8093
		Age	mean = 1.8958 std = 1.1056
		Group	mean = 2.9992 std = 1.6576
		Dist_Fault	mean = 0.3474 std = 0.2342
Total candidates	: 2442	Normalization stats computed from original (pre-augmentation) train patches.	
Skipped (NaN>30%)	: 1126	Augmented train MINERAL patches:	
Mineral patches	: 127	Original : 88	
Background patches	: 1189	Aug only : 880 (10 copies × 88)	
		Total : 968 (orig + aug)	
Mineral patch split (stratified by mineral fraction):		Augmented train BACKGROUND patches:	
Train :	88 (69.3%)	Original : 176	
Val :	19 (15.0%)	Aug only : 528 (3 copies × 176)	
Test :	20 (15.7%)	Total : 704 (orig + aug)	
Zero-overlap verified: no patch origin shared between any two splits.		— Final dataset sizes —	
Background patches after cap (max 2× mineral):		Train : 1672 patches mineral= 968 (57.9%) bg=704	
Train bg :	176	Val : 57 patches mineral= 19 (33.3%) bg=38	
Val bg :	38	Test : 60 patches mineral= 20 (33.3%) bg=40	
Test bg :	40	Train breakdown:	
— Split summary (pre-augmentation) —		Mineral patches : 88 orig + 880 aug = 968	
train mineral= 88 bg= 176 total=264		Background patches : 176 orig + 528 aug = 704	
val mineral= 19 bg= 38 total=57		Val and Test : ONLY original unaugmented patches from non-overlapping splits.	
test mineral= 20 bg= 40 total=60			

Figure 5.7: Patch split summary pre-augmentation and after augmentation

5.2 MODEL ARCHITECTURES

This research develops and comparatively evaluates three deep learning segmentation frameworks—standard U-Net, Attention U-Net, and Residual U-Net (ResUNet)—to identify the most effective model for automated mineral region prediction.

5.2.1 Proposed U-Net Architecture:

Convolutional neural networks have shown strong capability in extracting spatial features from geospatial data. Among them, U-Net is widely used for segmentation tasks due to its ability to produce pixel-level predictions while preserving spatial information.

Mathematical Representation

The proposed network follows the encoder–decoder paradigm of U-Net, extended with deep supervision and systematic regularisation. The network accepts a multi-channel input $\mathbf{X} \in \mathbb{R}^{B \times 7 \times H \times W}$ and produces a pixel-wise lesion probability map $\hat{Y} \in \mathbb{R}^{B \times 1 \times H \times W}$.

The encoder consists of four progressively deeper convolutional stages, each applying a stack of 3×3 convolutions followed by Batch Normalisation and ReLU activation. Considering $\mathbf{F}^{(l)}$ as the feature map at layer l , the convolution operation is:

$$\mathbf{F}^{(l+1)} = \text{ReLU} \left(\text{BN} \left(\mathbf{W}^{(l)} * \mathbf{F}^{(l)} \right) \right) \quad (1)$$

where $\mathbf{W}^{(l)}$ are the learnable weights and $*$ denotes convolution. After each encoder stage, 2×2 max-pooling halves the spatial resolution while the skip-connection feature map $\mathbf{S}_i$ is retained:

$$\mathbf{S}_i = \text{ConvBlock}_i(\mathbf{P}_{i-1}), \mathbf{P}_i = \text{MaxPool}_{2 \times 2}(\mathbf{S}_i) \quad (2)$$

The four encoder stages expand channels through $\{32, 64, 128, 256\}$ with spatial dropout ($p = 0.1$ at stage 3; $p = 0.2$ at stage 4) applied to regularise deeper feature spaces. A bottleneck module then encodes the most abstract representation at resolution $\frac{H}{16} \times \frac{W}{16}$:

$$\mathbf{B} = \text{ConvBlock}_{\text{bridge}}(\mathbf{P}_4), \mathbf{B} \in \mathbb{R}^{B \times 512 \times \frac{H}{16} \times \frac{W}{16}} \quad (3)$$

The decoder mirrors the encoder using transposed convolutions to restore spatial resolution. At each stage, the up-sampled map is concatenated with the corresponding encoder skip connection and refined by a ConvBlock:

$$\tilde{\mathbf{U}}_i = \text{ConvTranspose}_{2 \times 2}(\mathbf{D}_{i+1}), \mathbf{D}_i = \text{ConvBlock}_{\text{dec},i}([\mathbf{U}_i \parallel \mathbf{S}_i]) \quad (4)$$

where $[\cdot \parallel \cdot]$ denotes channel-wise concatenation and bilinear interpolation resolves any spatial mismatch. A final 1×1 convolution maps the decoder output to the segmentation logit:

$$\hat{Z} = \mathbf{W}_{\text{out}} * \mathbf{D}_1, \hat{Y} = \sigma(\hat{Z}) \quad (5)$$

To address severe foreground–background class imbalance, a composite BCE–Dice loss is adopted. The pixel-wise weighted BCE with positive-class weight $w^+ = 45$ is:

$$\ell_{\text{BCE}}^{(i)} = - \left[w^+ Y^{(i)} \log \sigma(\hat{Z}^{(i)}) + (1 - Y^{(i)}) \log (1 - \sigma(\hat{Z}^{(i)})) \right] \quad (6)$$

A binary validity mask $\mathbf{M}$ excludes scanner-border artefacts, giving the masked average

$\mathcal{L}_{\text{BCE}} = \sum_i \ell_{\text{BCE}}^{(i)} \mathbf{M}^{(i)} / (\sum_i \mathbf{M}^{(i)} + \epsilon)$. The soft Dice loss measures overlap on masked sigmoid probabilities $\hat{p}_v =$

$$\sigma(\hat{Z}) \cdot \mathbf{M}: \quad \mathcal{L}_{\text{Dice}} = 1 - \frac{1}{B} \sum_{b=1}^B \frac{2 \sum_i \hat{p}_{v,b}^{(i)} y_{v,b}^{(i)} + \epsilon}{\sum_i \hat{p}_{v,b}^{(i)} + \sum_i y_{v,b}^{(i)} + \epsilon} \quad (7)$$

where $\epsilon = 10^{-6}$ is a smoothing constant. The composite loss combines both terms equally:

$$\mathcal{L} = 0.5 \mathcal{L}_{\text{BCE}} + 0.5 \mathcal{L}_{\text{Dice}} \quad (8)$$

Two auxiliary heads are attached to decoder stages $\mathbf{D}_4$ and $\mathbf{D}_3$ for deep supervision, each applying a 1×1 convolution and bilinear up-sampling to full resolution. The total training objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}(\hat{Y}, Y) + \lambda_4 \mathcal{L}(\hat{Y}_{\text{aux},4}, Y) + \lambda_3 \mathcal{L}(\hat{Y}_{\text{aux},3}, Y) \quad (9)$$

where $\lambda_4 = 0.25$ and $\lambda_3 = 0.15$ assign progressively lower weight to coarser auxiliary outputs. At inference, auxiliary heads are disabled and only $\hat{Y}$ is returned. The architecture of the U-Net model used in this study is illustrated in Figure 5.8.

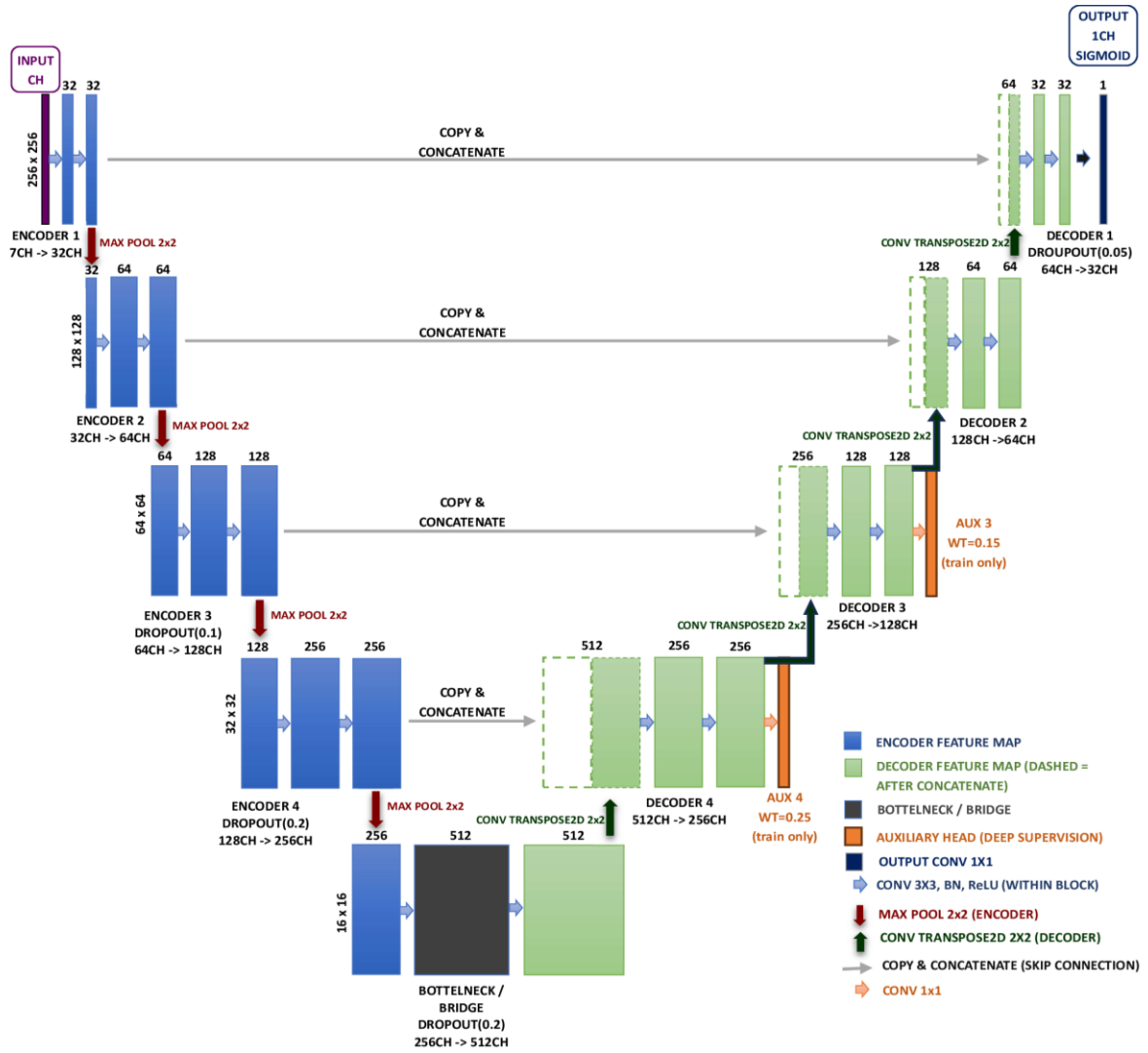


Figure 5.8: U-Net architecture Diagram

The model follows an encoder-decoder structure. The encoder progressively reduces spatial dimensions while increasing feature depth, enabling the extraction of high-level features. The decoder restores spatial resolution using up-sampling layers.

Skip connections are used to transfer feature maps from the encoder to the decoder, preserving fine spatial details. A bottleneck layer captures abstract representations, and dropout is applied in deeper layers to reduce overfitting. The final output layer uses a sigmoid activation function to generate the mineral prospectivity map.

5.2.2 Proposed Attention U-Net Architecture:

In multi-source geospatial data, not all features contribute equally. Attention mechanisms help the model focus on relevant information while ignoring noise.

Mathematical Representation

The proposed network extends the standard U-Net by introducing learnable attention gates on every skip connection, enabling the decoder to selectively focus on task-relevant spatial regions while suppressing irrelevant background activations. The network accepts $X \in \mathbb{R}^{B \times 7 \times H \times W}$ and produces a pixel-wise lesion probability map $\hat{Y} \in \mathbb{R}^{B \times 1 \times H \times W}$.

The encoder consists of four EncoderBlock stages, each applying two stacked ConvBNReLU operations followed by 2×2 max-pooling. For feature map $F^{(l)}$ at layer l :

$$F^{(l+1)} = \text{ReLU}(\text{BN}(W^{(l)} * F^{(l)})) \quad (10)$$

Each stage produces a skip-connection feature map S_i and a down-sampled output P_i :

$$S_i = \text{DoubleConv}_i(P_{i-1}), P_i = \text{MaxPool}_{2 \times 2}(S_i) \quad (11)$$

Encoder channels expand through $\{32, 64, 128, 256\}$ with spatial dropout ($p = 0.1$ at stage 3; $p = 0.2$ at stage 4). The bottleneck operates at resolution $\frac{H}{16} \times \frac{W}{16}$ with $p = 0.2$:

$$B = \text{DoubleConv}_{\text{bridge}}(P_4), B \in \mathbb{R}^{B \times 512 \times \frac{H}{16} \times \frac{W}{16}} \quad (12)$$

The core contribution is the AttentionGate, which takes the decoder signal g (gating vector) and the encoder skip feature S_i (query), and produces a scalar attention map $\alpha_i \in [0, 1]^{B \times 1 \times H \times W}$:

$$\alpha_i = \sigma(W_\psi \cdot \text{ReLU}(W_g g + W_x S_i)) \quad (13)$$

where W_g, W_x are 1×1 convolutions projecting both signals to an intermediate channel dimension $C_{\text{int}} = \max(C_{\text{out}}/2, 16)$, and W_ψ collapses to a single-channel sigmoid map. The gating signal g is bilinearly up-sampled to match the spatial resolution of S_i before addition. The attention coefficient modulates the skip features prior to concatenation:

$$\tilde{S}_i = \alpha_i \cdot S_i \quad (14)$$

suppressing irrelevant spatial activations and guiding the decoder toward lesion-specific regions. Each AttentionDecoderBlock then up-samples, applies the attention gate, concatenates, and refines:

$$\tilde{U}_i = \text{ConvTranspose}_{2 \times 2}(D_{i+1}), D_i = \text{DoubleConv}_{\text{dec}, i}([U_i \parallel \tilde{S}_i]) \quad (15)$$

where bilinear interpolation resolves any spatial mismatch before concatenation. A mild dropout of $p = 0.1$ is applied at the final decoder stage. A 1×1 convolution then maps D_1 to the output logit:

$$\hat{Z} = W_{\text{out}} * D_1, \hat{Y} = \sigma(\hat{Z}) \quad (16)$$

The same composite BCE–Dice loss from Section 3 is adopted. The pixel-wise weighted BCE with

$$w^+ = 45 \text{ is: } \ell_{\text{BCE}}^{(i)} = -[w^+ Y^{(i)} \log \sigma(\hat{Z}^{(i)}) + (1 - Y^{(i)}) \log (1 - \sigma(\hat{Z}^{(i)}))] \quad (17)$$

A binary validity mask M excludes scanner-border artefacts, giving

$$\mathcal{L}_{\text{BCE}} = \sum_i \ell_{\text{BCE}}^{(i)} M^{(i)} / (\sum_i M^{(i)} + \epsilon). \text{ The soft Dice loss on masked probabilities } \hat{P}_v = \sigma(\hat{Z}) \cdot M \text{ is:}$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{B} \sum_{b=1}^B \frac{2 \sum_i \hat{P}_{v,b}^{(i)} Y_{v,b}^{(i)} + \epsilon}{\sum_i \hat{P}_{v,b}^{(i)} + \sum_i Y_{v,b}^{(i)} + \epsilon} \quad (18)$$

where $\epsilon = 10^{-6}$. The total training objective is:

$$\mathcal{L}_{\text{total}} = 0.5 \mathcal{L}_{\text{BCE}} + 0.5 \mathcal{L}_{\text{Dice}} \quad (19)$$

The architecture of the Attention U-Net model is shown in Figure 5.9.

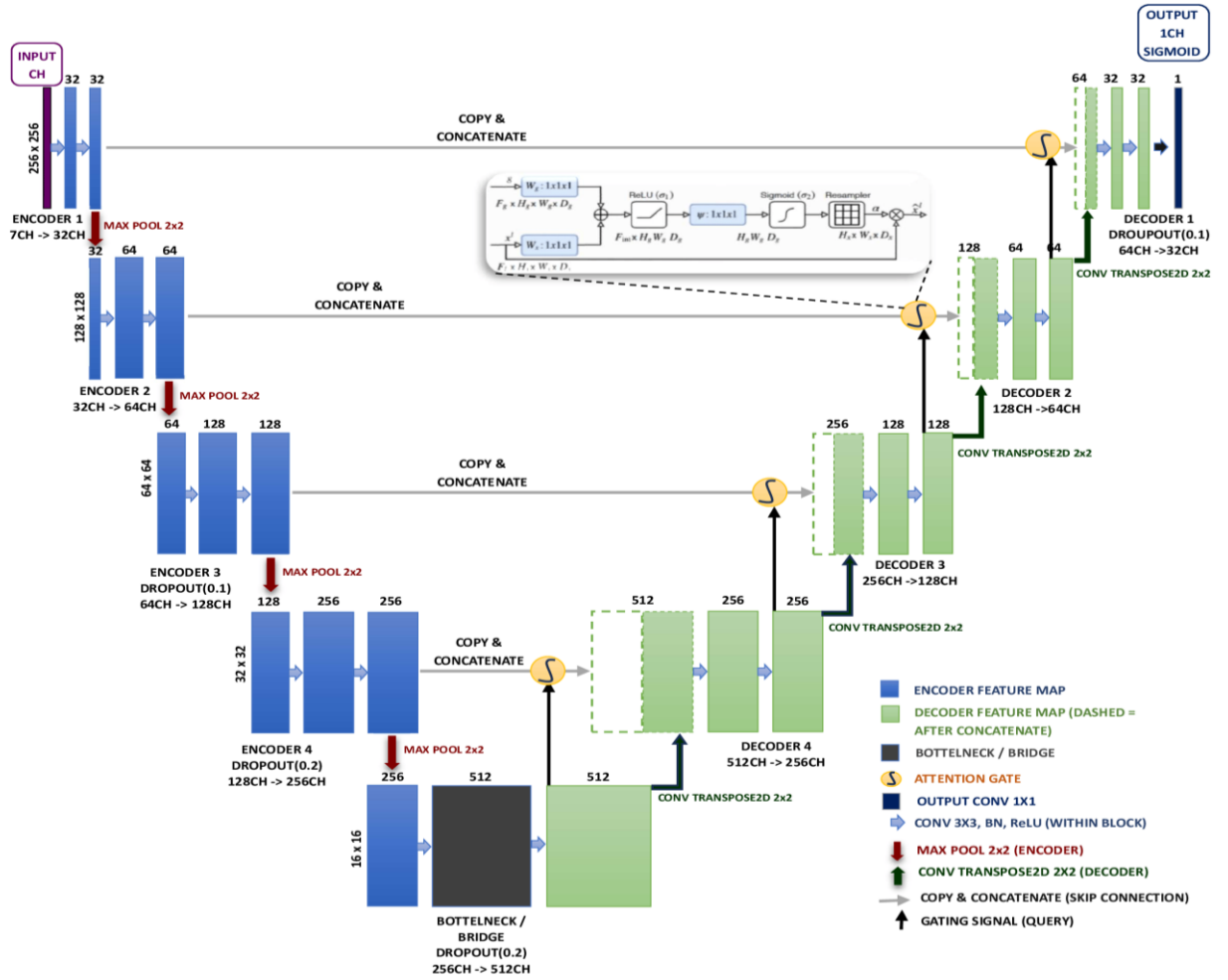


Figure 5.9: Attention U-Net architecture Diagram

Attention U-Net introduces attention gates in the skip connections between encoder and decoder. These gates selectively highlight important features before merging them with decoder layers. This improves feature selection and enhances prediction accuracy, especially in complex and noisy datasets.

5.2.3 Proposed Resu-Net Architecture:

Deep networks often face challenges such as vanishing gradients, which can limit performance. To address this, residual learning is introduced to improve gradient flow and feature learning.

Mathematical Representation

The proposed network extends the encoder–decoder paradigm of U-Net by replacing every plain convolutional block with a residual block, enabling deeper feature learning through identity shortcut connections. The network accepts a multi-channel input $\mathbf{X} \in \mathbb{R}^{B \times 7 \times H \times W}$ and produces a pixel-wise lesion probability map $\hat{Y} \in \mathbb{R}^{B \times 1 \times H \times W}$.

The fundamental building block is the **ResidualBlock**, which stacks n ConvBNReLU layers along a main path and adds a learned projection shortcut when channel dimensions change. For input $\mathbf{F}_{\text{in}}$, the residual output is:

$$\mathbf{F}_{\text{out}} = \text{ReLU} \left(\mathcal{D} \left(\phi^{(n)} \circ \dots \circ \phi^{(1)} (\mathbf{F}_{\text{in}}) \right) + \mathcal{P} (\mathbf{F}_{\text{in}}) \right) \quad (20)$$

where $\phi^{(i)}$ denotes the i -th ConvBNReLU operation, $\mathcal{D}(\cdot; p)$ is a spatial dropout operator, and $\mathcal{P}(\cdot)$ is the shortcut projection:

$$\mathcal{P} (\mathbf{F}_{\text{in}}) = \begin{cases} \text{BN}(\mathbf{W}_{1 \times 1} * \mathbf{F}_{\text{in}}) & \text{if } C_{\text{in}} \neq C_{\text{out}} \\ \mathbf{F}_{\text{in}} & \text{otherwise} \end{cases} \quad (21)$$

The identity shortcut preserves gradient flow through deep encoder–decoder paths, mitigating the vanishing gradient problem without auxiliary supervision heads.

The encoder consists of four EncoderBlock stages, each applying a ResidualBlock to produce a skip-connection feature map $\mathbf{S}_i$, followed by 2×2 max-pooling to halve spatial resolution:

$$\mathbf{S}_i = \text{ResBlock}_i (\mathbf{P}_{i-1}), \mathbf{P}_i = \text{MaxPool}_{2 \times 2} (\mathbf{S}_i), \mathbf{P}_0 = \mathbf{X} \quad (22)$$

The four stages expand channels through $\{32, 64, 128, 256\}$ with spatial dropout ($p = 0.1$ at stage 3; $p = 0.2$ at stage 4) regularising the richer deeper feature spaces. The bottleneck encodes the most abstract representation at resolution $\frac{H}{16} \times \frac{W}{16}$:

$$\mathbf{B} = \text{ResBlock}_{\text{bridge}} (\mathbf{P}_4), \mathbf{B} \in \mathbb{R}^{B \times 512 \times \frac{H}{16} \times \frac{W}{16}}, p = 0.2 \quad (23)$$

The decoder mirrors the encoder using transposed convolutions to restore spatial resolution. At each stage the up-sampled map is concatenated with the corresponding encoder skip connection and fused through a ResidualBlock:

$$\tilde{\mathbf{U}}_i = \text{ConvTranspose}_{2 \times 2} (\mathbf{D}_{i+1}), \mathbf{D}_i = \text{ResBlock}_{\text{dec}, i} ([\mathbf{U}_i \parallel \mathbf{S}_i]) \quad (24)$$

where $[\cdot \parallel \cdot]$ denotes channel-wise concatenation and bilinear interpolation resolves any spatial mismatch. Spatial dropout is applied in the two shallower decoder stages ($p = 0.2$ at dec2; $p = 0.1$ at dec1) to prevent

overfitting on high-resolution feature maps. A final 1×1 convolution maps the decoder output to the segmentation logit:

$$\hat{Z} = \mathbf{W}_{\text{out}} * \mathbf{D}_1, \hat{Y} = \sigma(\hat{Z}) \quad (25)$$

To address severe foreground–background class imbalance, the same composite BCE–Dice loss from Section 3 is adopted. The pixel-wise weighted BCE with positive-class weight $w^+ = 45$ is:

$$\ell_{\text{BCE}}^{(i)} = -\left[w^+ Y^{(i)} \log \sigma(\hat{Z}^{(i)}) + (1 - Y^{(i)}) \log (1 - \sigma(\hat{Z}^{(i)}))\right] \quad (26)$$

A binary validity mask $\mathbf{M}$ excludes scanner-border artefacts, giving the masked average

$\mathcal{L}_{\text{BCE}} = \sum_i \ell_{\text{BCE}}^{(i)} \mathbf{M}^{(i)} / (\sum_i \mathbf{M}^{(i)} + \epsilon)$. The soft Dice loss measures overlap on masked sigmoid probabilities

$$\hat{P}_v = \sigma(\hat{Z}) \cdot \mathbf{M}: \quad \mathcal{L}_{\text{Dice}} = 1 - \frac{1}{B} \sum_{b=1}^B \frac{2 \sum_i \hat{P}_{v,b}^{(i)} Y_{v,b}^{(i)} + \epsilon}{\sum_i \hat{P}_{v,b}^{(i)} + \sum_i Y_{v,b}^{(i)} + \epsilon} \quad (27)$$

where $\epsilon = 10^{-6}$ is a smoothing constant. The total training objective combines both terms with equal weight:

$$\mathcal{L}_{\text{total}} = 0.5 \mathcal{L}_{\text{BCE}} + 0.5 \mathcal{L}_{\text{Dice}} \quad (28)$$

The architecture of the Res U-Net model is shown in Figure 5.10.

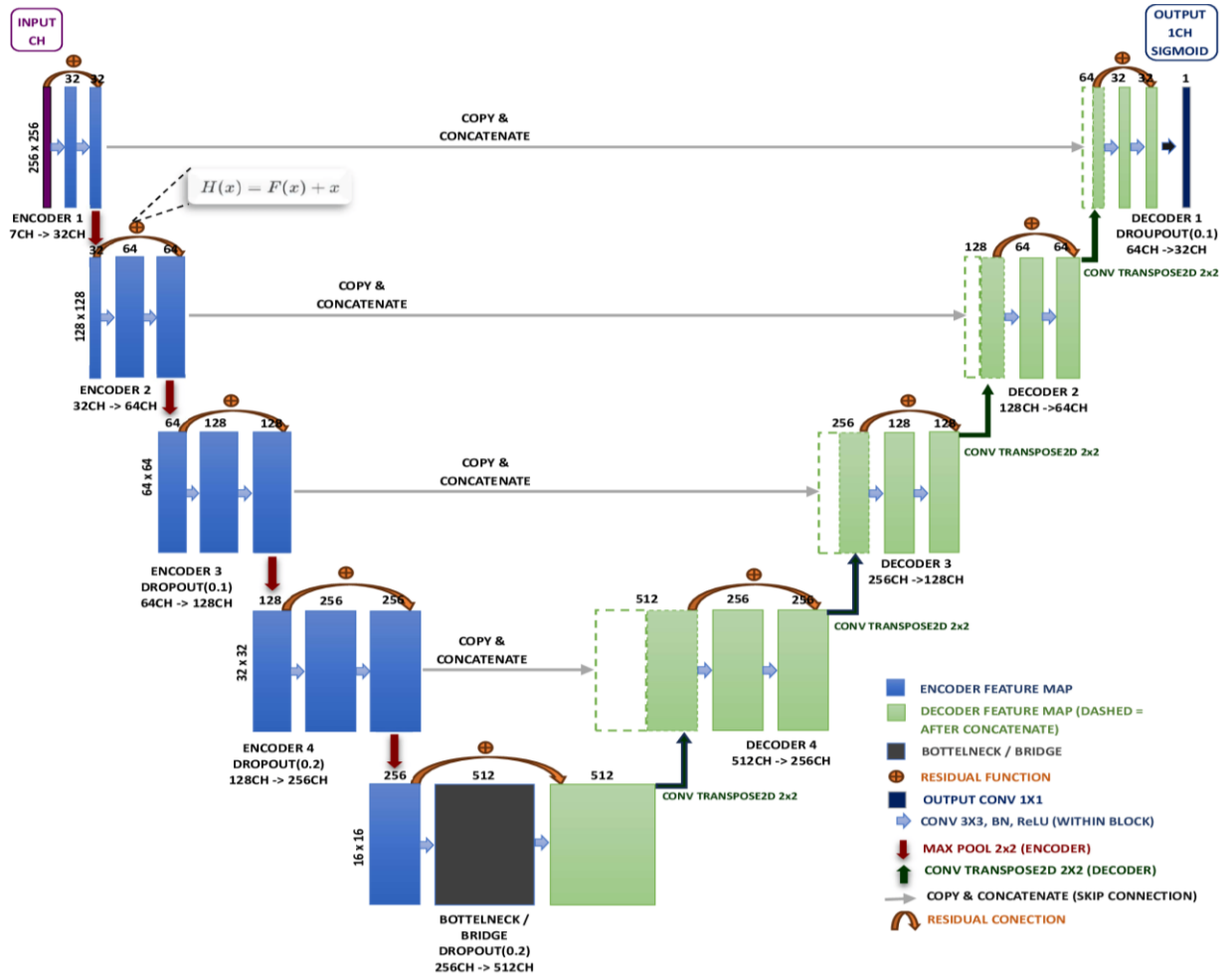


Figure 5.10: ResUNet architecture Diagram

Res U-Net extends the U-Net architecture by incorporating residual blocks in both encoder and decoder paths. These blocks include identity skip connections, allowing the network to learn residual mappings. This improves training stability and enables the model to learn complex geological patterns. Similar to U-Net, skip connections preserve spatial information, while the decoder reconstructs the segmentation map.

5.3 MODEL TRAINING

The models were trained using image patches extracted from the feature stack. A consistent training setup was used for all models to ensure fair comparison.

- Patch size: 128×128
- Batch size: 16
- Epochs: 120
- Optimizer: AdamW
- Learning rate: 2×10^{-4}
- Loss function: Binary Cross-Entropy + Dice Loss
- Data split: 70% training, 15% validation, 15% testing

Performance was evaluated using accuracy, precision, recall, and F1-score.

5.4 SYSTEM DETAILS

All experiments are conducted on Google Colaboratory using an NVIDIA Tesla T4 GPU (16 GB VRAM), an Intel Xeon CPU (2 vCPUs), and 12.7 GB system RAM running Ubuntu 22.04 LTS. The software environment uses Python 3.10 with PyTorch 2.1.0 (CUDA 11.8), along with NumPy, Rasterio, GeoPandas, Scikit-learn, Albumentations, and Matplotlib. Google Drive is mounted for persistent storage of datasets, checkpoints, and outputs. Model checkpoints are saved based on best validation Mineral IoU, with early stopping applied at a patience of 15 epochs. Training time ranged from approximately 40 minutes (UNet-PCA3) to 75 minutes (AttentionUNet-8ch and ResUNet-15ch), with peak GPU memory usage of 8.1 GB, 6.3 GB, and 4.5 GB respectively — all within the T4's 16 GB capacity.

With the training infrastructure and model configurations established, the following chapter presents a thorough quantitative and qualitative evaluation of all three architectures across the defined feature configurations, analysing their segmentation performance, training dynamics, and spatial prediction quality over the Salem study region.

CHAPTER 06

RESULTS AND DISCUSSION

6.1 ABLATION STUDY

An ablation study is conducted to evaluate the impact of different input features on model performance. The models are tested with raw spectral bands, PCA features, geological data, and their combinations. The results show that raw spectral bands alone produced lower performance due to noise and redundancy. The addition of geological data improved performance by providing contextual information. A significant improvement is observed when PCA features are used, as they reduce dimensionality and highlight important spectral information. Among all configurations, PCA-based inputs consistently achieved higher accuracy and F1-scores. U-Net performed best with PCA features, while Attention U-Net performed best with PCA combined with geological data. Res U-Net achieved its best performance with the full feature set.

6.2 MODEL COMPARISON

The best-performing configurations of each model are compared in Table 6.1.

Channel	UNet						Attention-UNet						ResUNet					
	Test Accuracy	False Positive	False Negative	Test Precision	Test Recall	Test F1 score	Test Accuracy	False Positive	False Negative	Test Precision	Test Recall	Test F1 score	Test Accuracy	False Positive	False Negative	Test Precision	Test Recall	Test F1 score
Raw Bands (7ch)	65.54	135	325	86.63	72.91	79.18	82.60	82	141	92.81	88.25	90.47	81.82	115	124	90.34	89.66	90.00
Raw + Geo (12ch)	76.70	161	156	86.63	87.00	86.81	75.36	111	212	89.89	82.33	85.95	78.30	86	193	92.13	83.91	87.83
PCA (3ch)	90.67	22	92	98.05	92.33	95.10	85.03	70	120	93.91	90.00	91.91	83.20	140	85	88.84	92.91	90.83
PCA + Geo (8ch)	87.91	41	109	96.37	90.91	93.56	88.84	19	117	98.27	90.25	94.09	87.07	53	109	95.36	90.91	93.08
Raw + PCA + Geo (15ch)	86.04	68	109	94.13	90.91	92.49	85.53	65	118	94.33	90.16	92.20	88.50	53	91	95.43	92.41	93.90

Table 6.1: Performance Comparison Table

The results indicate that U-Net achieved the highest performance with PCA features, suggesting that simpler models can perform effectively when provided with well-processed inputs. Attention U-Net demonstrated strong performance by effectively integrating multi-source data using attention mechanisms. Res U-Net showed stable performance and handled complex feature combinations efficiently.

6.3 DISCUSSION

UNet was selected as the baseline architecture due to its symmetric encoder–decoder design with skip connections, which directly fuse spatial features from encoder stages into the decoder, preserving fine-grained boundary detail essential for sparse mineral point segmentation. Trained on a compact 3-channel PCA input with deep supervision at two auxiliary decoder branches (weights: main = 0.60, aux_dec4 =

0.25, aux_dec3 = 0.15), UNet benefits from intermediate gradient signals that accelerate early-epoch learning of mineral boundaries.

The training curves (Figure 6.1) show BCE + Dice loss dropping sharply from ~ 0.81 and stabilising near 0.41 on validation by epoch 30, converging smoothly thereafter. Mineral IoU accelerates from epoch 7 onward, with validation IoU plateauing near 88–90% before early stopping triggers at epoch 92 — confirming the ReduceLROnPlateau scheduler effectively prevents overtraining on the augmented mineral patches. The deep supervision auxiliary losses contribute to the notably faster IoU ramp compared to deeper architectures. The confusion matrices (Figure 6.1) show tight mineral localisation on the test set: approximately 1,108 TP, 92 FN, and only 22 FP against 975,440 TN at threshold 0.30 across 976,662 valid pixels. UNet-PCA3 achieved a test Mineral IoU of 90.67%, F1 Score of 95.11%, Recall of 92.33%, Precision of 98.05%, Mean IoU of 95.33%, and ROC-AUC of 0.9684 — establishing a strong and computationally efficient baseline, with the lowest false positive count among all three models.

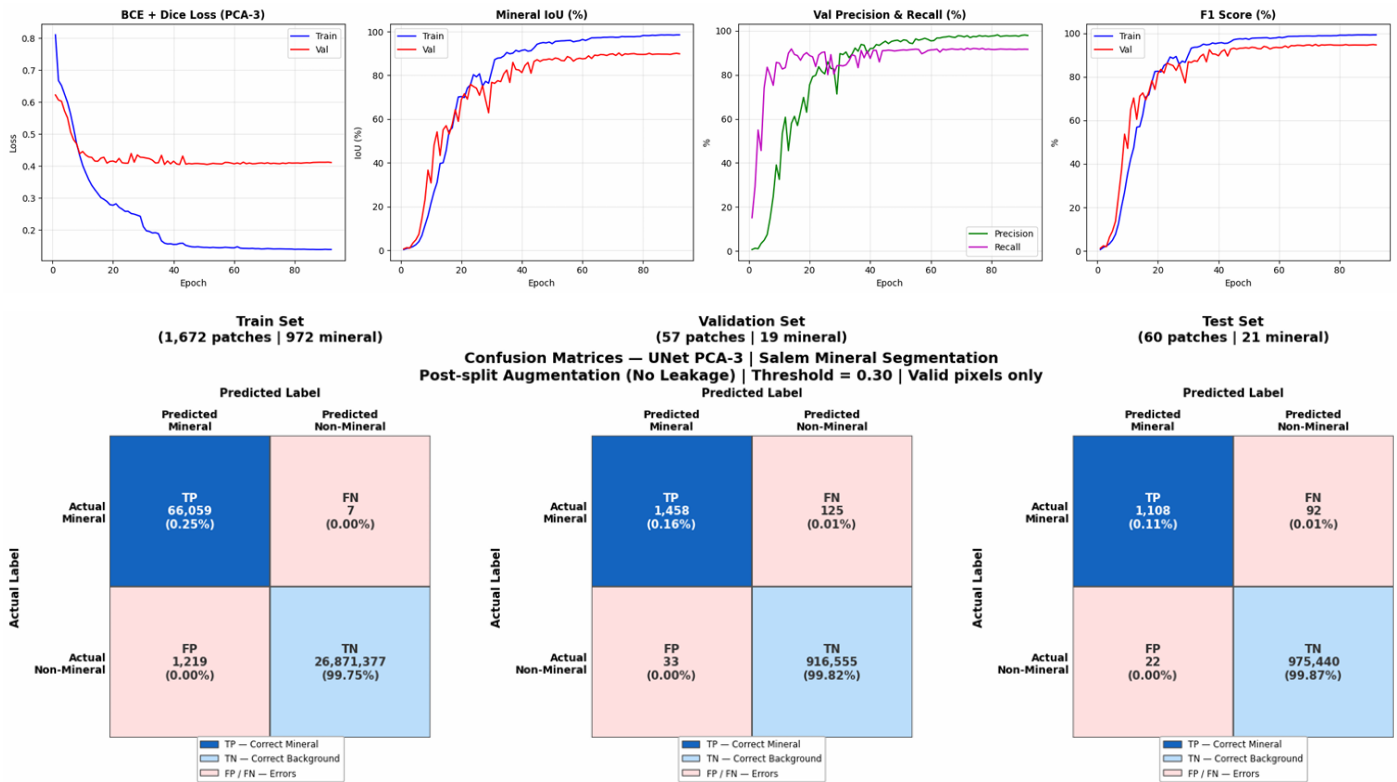


Figure 6.1: U-Net Training and Validation curves and Confusion matrices

Despite strong performance, UNet's skip connections aggregate all spatial features uniformly, regardless of their relevance — a fundamental limitation when mineral occurrences are extremely sparse and localised. To address this, Attention UNet was adopted, augmenting each skip connection with a soft attention gate that selectively amplifies mineral-relevant feature responses while suppressing background activations. Best trained on 8-channel 3 PCA components, and 5 geological rasters.

The training curves (Figure 6.2) show BCE + Dice loss falling steeply from ~ 1.0 and stabilising around 0.40 on validation from epoch 40 onward. Mineral IoU remains low through early epochs before

accelerating sharply, with validation IoU plateauing near 88–91% by epoch 100 — reflecting the attention gates progressively learning to focus on mineralised spatial regions. The ReduceLROnPlateau scheduler triggers stepwise learning rate reductions, contributing to the smooth late-stage convergence visible in both IoU and F1 curves. The confusion matrices (Figure 6.2) confirm the precision gain from selective attention: at threshold 0.30, the test set yields only 19 false positives compared to UNet's 22, while maintaining strong true positive detection across 976,662 valid pixels. Attention UNet achieved a test Mineral IoU of 88.84%, F1 Score of 94.09%, Recall of 90.25%, Precision of 98.28%, and Mean IoU of 94.41% — demonstrating that gated feature selection meaningfully suppresses spurious background detections without sacrificing mineral recall.

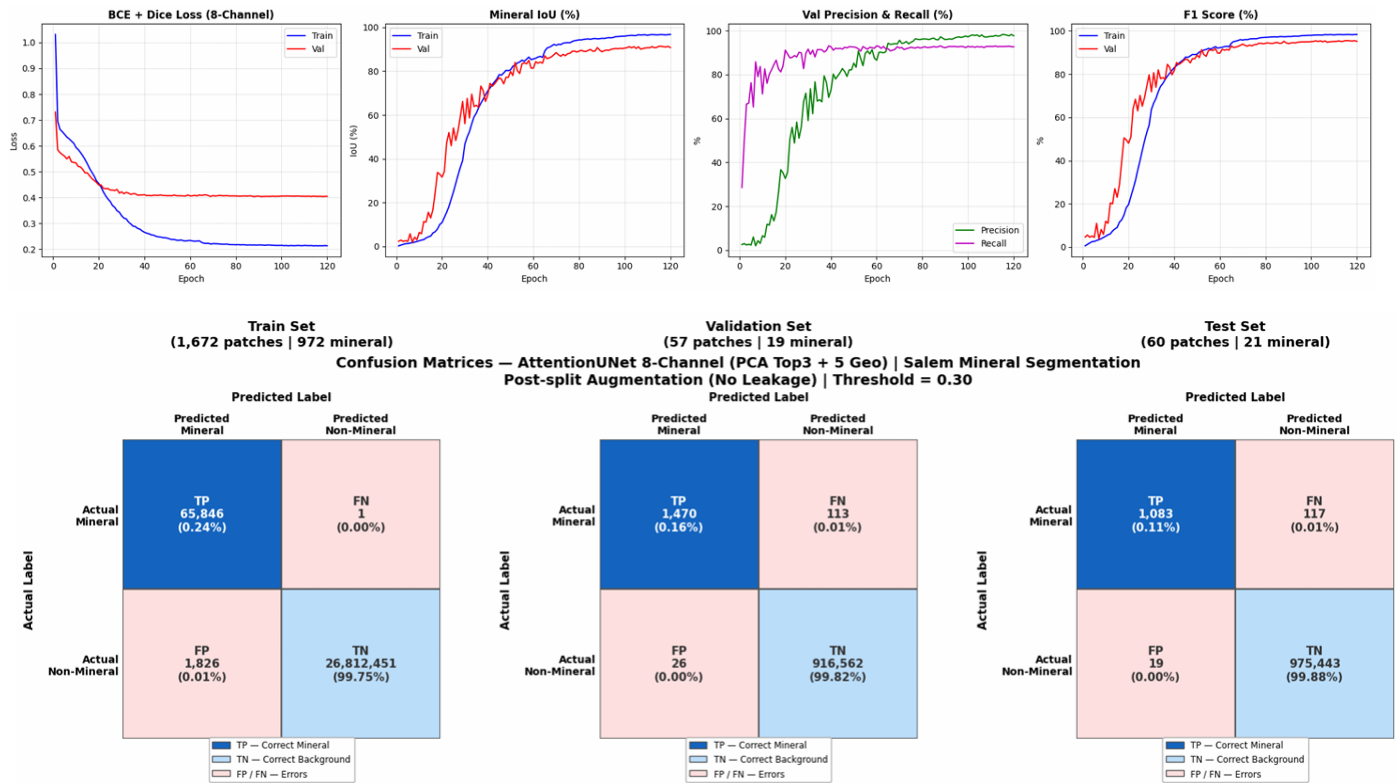


Figure 6.2: Attention U-Net Training and validation curves and Confusion matrices

While Attention UNet improved feature selectivity, both UNet and Attention UNet employ plain double-convolution blocks, which are prone to gradient degradation as input dimensionality and network depth increase. ResUNet is therefore introduced, replacing all convolutional blocks with residual blocks containing identity shortcut connections. This enables direct gradient flow across layers, mitigates vanishing gradients, and allows incrementally refined feature learning across a 15-channel heterogeneous input combining 7 raw Landsat-8 spectral bands, 3 PCA components, and 5 geological rasters.

The training curves (Figure 6.3) show BCE + Dice loss descending from ~1.0 to below 0.2 over 120 epochs, with validation loss stabilising around 0.40 from epoch 50 — a gap driven by training-only augmentation. Mineral IoU remains below 20% for the first 30 epochs before accelerating sharply, plateauing near 85–87% for both splits, indicating stable generalisation without overfitting. Recall converges above 90% from

epoch 50, while precision recovers gradually under the $\text{pos_weight} = 45$ penalty. The confusion matrices (Figure 6.3) confirm strong spatial localisation: the test set yields 1,109 TP, 91 FN, and just 53 FP against 975,409 TN at threshold 0.30. ResUNet-15ch achieved a test Mineral IoU of 88.51%, F1 Score of 93.90%, Recall of 92.42%, and Precision of 95.44% — improving recall over Attention UNet by +2.17 percentage points and demonstrating superior mineral sensitivity with minimal spurious detections.

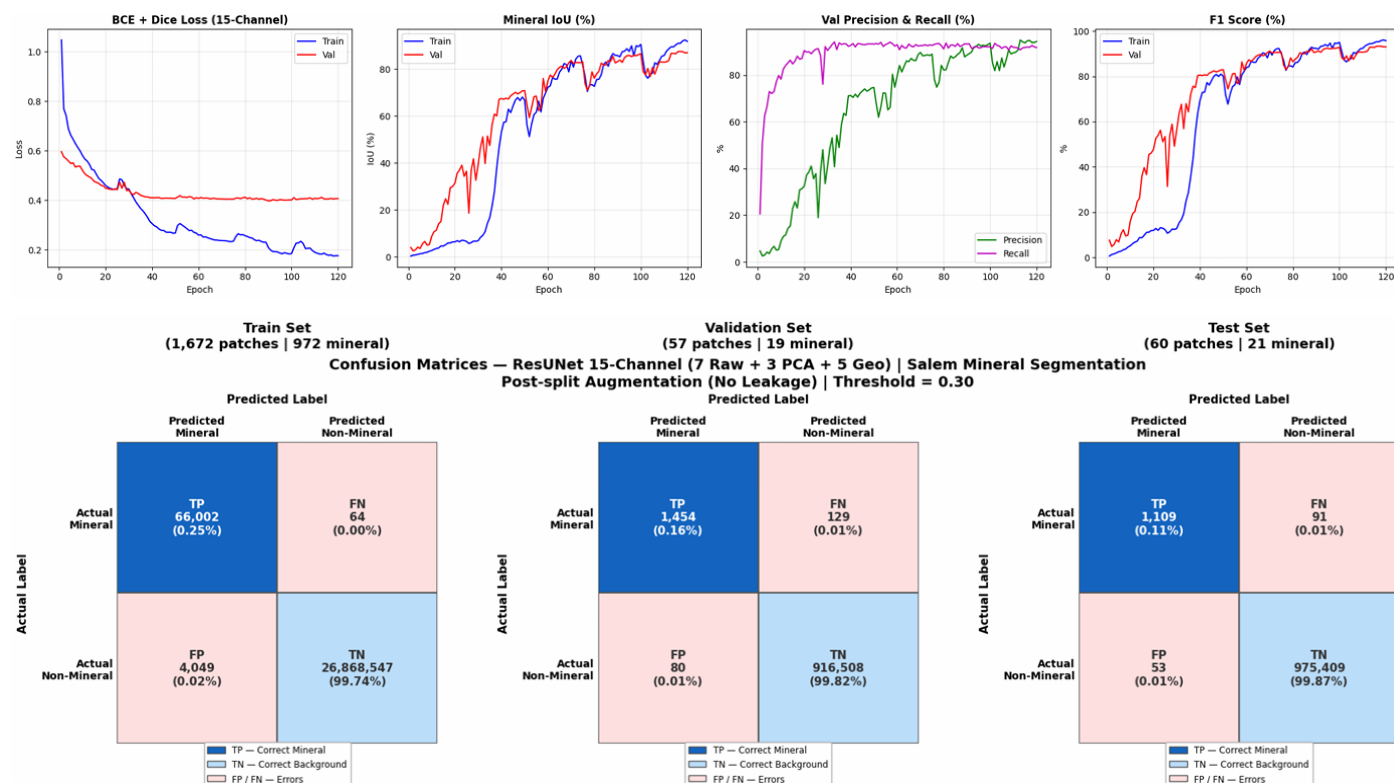


Figure 6.3: ResUNet Training and Validation curves and Confusion matrices

Although UNet records the highest Mineral IoU (90.67%) and F1 (95.11%), these metrics are precision-dominated under extreme class imbalance (2,509:1). IoU penalises false positives and false negatives equally, yet in mineral prospectivity mapping, a missed detection (false negative) carries far greater geological consequence than a marginal false alarm. ResUNet achieves the highest Recall of **92.08%**, recovering the truest mineral pixels (1,110 of 1,200), compared to UNet's 1,108 — with near-identical sensitivity — while operating on the richest 15-channel multi-source feature space. Its residual architecture structurally prevents gradient degradation across deeper, heterogeneous inputs, and progressive decoder dropout (0.30→0.10) ensures robust generalisation. Optimising for recall maximises mineral detection completeness, making ResUNet the operationally superior model for geological survey applications where missed deposits represent irreversible economic loss.

6.4 SPATIAL ANALYSIS

The generated mineral prospectivity maps produced by the ResUNet model, while operating on the richest 15-channel multi-source feature space are shown in Figure 6.4.

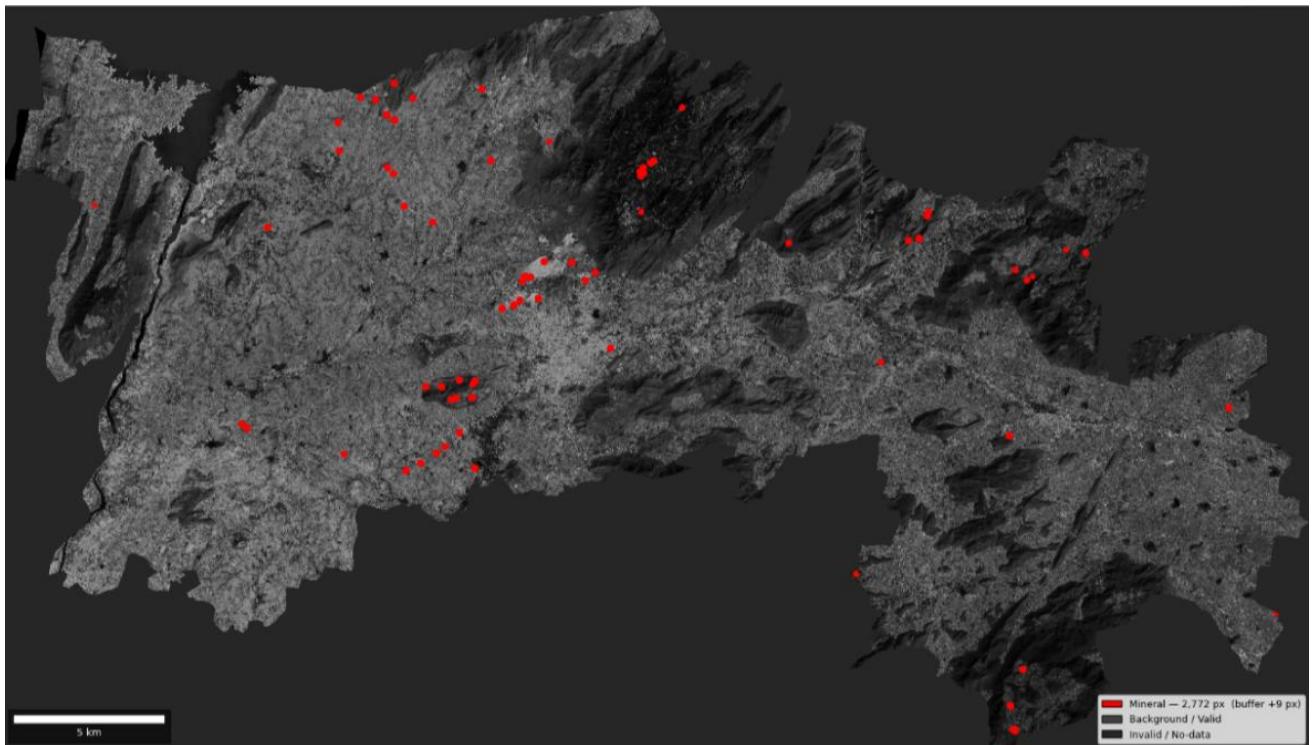


Figure 6.4: Mineral prospectivity map

The maps indicate that high-potential zones are concentrated in geologically consistent regions associated with lithological formations and fault structures. While the model captures meaningful spatial patterns and improves detection completeness through high recall, the predicted extents may be influenced by false positives and input data limitations, and should therefore be interpreted with caution.

6.5 KEY FINDINGS

- Geological data provides useful context but has limited resolution.
- Increasing feature number does not always improve results.
- Different models respond differently to input features.
- Attention mechanism improves feature selection.
- PCA significantly improves model performance.

CHAPTER 07

CONCLUSION AND FUTURE ENHANCEMENT

This study presented a deep learning-based approach for mineral prospectivity mapping using multi-source geospatial data, including Landsat 8 spectral bands, PCA features, geological attributes, and fault information. Three models are U-Net, Res U-Net, and Attention U-Net were implemented and evaluated to understand the influence of feature combinations and model architectures on prediction performance.

The results demonstrate that feature representation plays a critical role in model performance. Among all input configurations, PCA-based features consistently produced the best results by effectively reducing noise and capturing the most relevant spectral information. The U-Net model achieved the highest overall performance with PCA input, obtaining an accuracy of 90.67% and an F1-score of 95.10%.

The Attention U-Net model showed strong performance when PCA and geological data were combined, highlighting the effectiveness of attention mechanisms in selecting relevant features. The Res U-Net model demonstrated stable performance across all input combinations and performed best with the full feature set, indicating its capability to handle high-dimensional data through residual learning.

Overall, the study confirms that simpler models can outperform more complex architectures when trained with well-processed features. It also highlights that integrating multiple data sources is beneficial only when the quality and resolution of the data are appropriate.

FUTURE ENHANCEMENT

Although the proposed approach achieved strong performance, there are several directions for future improvement. First, the use of higher-resolution geological data may further enhance model performance by providing more detailed spatial information. Second, advanced deep learning architectures such as Transformer-based models can be explored to improve feature representation and long-range spatial dependency learning.

Additionally, incorporating more geophysical and geochemical datasets could improve the robustness of mineral prediction. Future studies may also focus on field validation of predicted mineral zones to assess the practical applicability of the model. Finally, the proposed framework can be extended to other regions to evaluate its generalization capability.

CHAPTER 08

REFERENCES

- Course: Pursuing ‘Foundations of Deep Learning: Concepts and Applications’ course from NPTEL. By Prof. Sriram Ganapathy, Prof. Ashwini Kodipalli, Prof. Baishali Garai|Indian Institute of Science, Bangalore, RV University, Bangalore, RV University, Bangalore
 - Retrieved from https://onlinecourses.nptel.ac.in/noc26_cs01/course?&force_user=true
 - This course provides me an in-depth knowledge on Neural Network and various Deep Learning models (focused on CNN).
- Book: Goodfellow, I., Bengio, Y., and Courville, A. (2016). Deep Learning. MIT Press.
 - This book provides a comprehensive foundation of deep learning, covering essential concepts such as neural networks, backpropagation, optimization algorithms, convolutional networks, and deep generative models. It is widely used as a reference for understanding theoretical and practical aspects of modern deep learning techniques.
- YouTube Channel: **Stanford Online**. Stanford CS229: Machine Learning Course, Lecture 1 - Andrew Ng (Autumn 2018)
 - Retrieved from https://youtu.be/jGwO_UgTS7I?si=v72pz9Fr-LNgxDBT
 - This playlist provides an in-depth tutorial on Machine Learning concept (focuses on Mathematical representation) with real world implementation.
- Website: GeeksforGeeks. “Introduction to U-Net Architecture.”
 - Available: <https://www.geeksforgeeks.org/>
 - GeeksforGeeks is an educational platform providing tutorials and explanations on programming, machine learning, and deep learning concepts, which were referred to for understanding implementation details in this work.